

VFSTR - VADLAMUDI	Regd. No.										
	Year	Semester			Branch			Section			
Staff Name: Dr. P. Srinivasa Rao	I	2			CSE, ECE, ECEVLSI			1 to 25			
Faculty Dept : Physics	Course: EP			Code: 25PY101			Set-1				
Program Name: B. Tech.	Date: 28.03.26			Time: 1:00 to 2.30 PM			T1				

Time: 90 Minutes

Max. Marks: 30

Part-A

Answer all questions

1. What are matter waves? Write the characteristics of matter waves? **5M**
2. a) State Heisenberg's uncertainty principle? Write the mathematical expression of Heisenberg's uncertainty principle. **3M**
b) An electron has an uncertainty in velocity of 5×10^5 m/s. Find the uncertainty in its position. (Mass of electron = 9.1×10^{-31} kg). **2M**

Part-B

Answer all questions

- a) Apply the time independent Schrödinger wave equation, discuss the solution for a particle in one dimensional potential well of infinite height. Hence obtain the normalized wave function. **8M**
b) An electron possesses a wavelength of $3\text{\AA}$ in the ground state of the potential well. Calculate the length of the potential well. **2M**
- d) a) Apply the Newton's law $F=ma$ for a free electron in a conductor. Obtain an expression for the current density in terms of electrical conductivity and electric field **8M**
b) Calculate the acceleration of a free electron in an electric field gradient of 200V/m . **2M**